

[illegible]

INA240A1: gain 20
Shunt resistor: 4mOhm/2 (parallel)
Cumulative Gain: 0.002*20 = 0.04 V/A
Range: 3.3/2/0.04 = +-41A

The left diagram shows the INA240A1 circuit with a shunt resistor (RSH) connected to the IN+ and IN- pins. The output is labeled C2_OUT. The right diagram shows the INA240A1 circuit with a shunt resistor (RSH) connected to the IN+ and IN- pins. The output is labeled C1_OUT. Both diagrams include a bypass capacitor (CBYP4 or CBYP3) connected to the GND pin.

L7808 Circuit + Enable/Disable config

VDS

limiting the voltage to limit the current
max current 50Amps ($I_{L_OCP} = 40s \text{ impulse}$)
drain-source resistance 4mOhm
 $V_{DS} = 50A \cdot 0.004Ohm = 0.2V$

configuration

$V_{DS} = \text{Tied to AGND } 0.06V$
 $V_{DS} = 18 \text{ k}\Omega \pm 5\% \text{ tied to AGND } 0.13V$
 $V_{DS} = 75 \text{ k}\Omega \pm 5\% \text{ tied to AGND } 0.26V \text{ (BEST)}$
 $V_{DS} = Hi-Z, 0.6V$
 $V_{DS} = 75 \text{ k}\Omega \pm 5\% \text{ tied to DVDD } 1.13VV$
 $V_{DS} = 18 \text{ k}\Omega \pm 5\% \text{ tied to DVDD } 1.88$

IDRIVE

Setting the target gate current
directly determining the mosfet rise time
our mosfet gate-source charge $Q = 3nC$
rise time = $Q/IDRIVE$ (pp 66)
we want the rise time of 50ns
 $IDRIVE = Q/50ns = 0.06A$

configuration

- IDRIVE = Tied to AGND 10mA
- IDRIVE = 18 kΩ ± 5% tied to AGND 30mA
- IDRIVE = 75 kΩ ± 5% tied to AGND 30mA (BEST)
- IDRIVE = Hi-Z 120mA
- IDRIVE = 75 kΩ ± 5% tied to DVDD 260mA
- IDRIVE = 18 kΩ ± 5% tied to DVDD 570mA
- IDRIVE = Tied to DVDD 1000mA

MODE

Setting the 3xPWM mode
Option for 6xPWM as well
pp 31

configuration

MODE - 00b = Tied to AGND 6xPWM
MODE - 01b = 47 kΩ ± 5% tied to AGND 3xPWM

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The diagram shows a circuit connection for the MODE pin. A green line labeled 'MODE' enters a blue rectangular component. From the right side of the component, a red zigzag line representing a resistor labeled 'R10' with a value of '47k' extends to the right. This resistor is then connected to a green line that terminates in a ground symbol labeled 'GND'.

30V max
FSamp max
 $R_{on} = 4m\Omega$
 $Q_{gd} = 3nC$

Legend: R/C (blue box) connected to GND.

Diagram 1 (Left): HV_IN pin configuration for GHC and GLC channels. The circuit shows two MOSFETs (UC1, UC2) with gates connected to HV_IN and sources connected to GND. The drains are connected to GND through capacitors CB1 and CB2. The output nodes are labeled SHC and SHC.

Diagram 2 (Middle): HV_IN pin configuration for GHB and GLB channels. The circuit shows two MOSFETs (UB1, UB2) with gates connected to HV_IN and sources connected to GND. The drains are connected to GND through capacitors CB2 and CB2. The output nodes are labeled SHB and SHB.

Diagram 3 (Right): HV_IN pin configuration for GHA and GLA channels. The circuit shows two MOSFETs (UA1, UA2) with gates connected to HV_IN and sources connected to GND. The drains are connected to GND through capacitors CB3 and CB3. The output nodes are labeled SHA and SHA.

ENABLE

R11
4.7kΩ

GND

 OUT
 pinA0

 OUT
 pinA1

 OUT
 pinA2

 OUT
 pinA3

The diagram shows a green line representing the VDD signal. It starts at a low level, then steps up to 3.3V, and then steps up again to 5V. The transition from 3.3V to 5V is labeled with a green arrow and the text "VDD".

Encoder/Hall

I2C

The diagram illustrates three parallel branches connected between a common 5V source and ground. Each branch contains a 5V source, a 10k resistor, and a diode in series. The diodes are labeled M1, M2, and M3. The connections are as follows:

- Branch 1: 5V source connected to a 10k resistor, which is connected to the anode of diode M1. The cathode of M1 is connected to ground.
- Branch 2: 5V source connected to a 10k resistor, which is connected to the anode of diode M2. The cathode of M2 is connected to ground.
- Branch 3: 5V source connected to a 10k resistor, which is connected to the anode of diode M3. The cathode of M3 is connected to ground.

Encoder/Hall Interface

SPI/I2C jst

The diagram illustrates the wiring for the Encoder/Hall Interface. It features two JST connectors, CN1 and CN2, connected to a microcontroller. CN1 (SM06B-SRSS-TB0,F(J)SN) has pins 1 (GND), 2 (VDD), 3 (GND), 4 (ENC_B), 5 (ENC_A), and 6 (GND). CN2 (SM04B-SRSS-TB0,F(SN)P) has pins 1 (GND), 2 (VDD), 3 (GND), 4 (SD), 5 (SD), and 6 (GND). The microcontroller pins are labeled P_ENC1, Header-Female-2.54_1x5, and ENC_A, ENC_B, ENC_C.

The diagram illustrates the wiring for an Arduino Header R3 Encoder/Hall Interface. It shows the following connections:

- Power Connections:**
 - 3.3V is connected to pin 1 of the P4 Header-Female 2.54_1x8.
 - +5V is connected to pin 2 of the P4 Header-Female 2.54_1x8.
 - VIN is connected to pin 4 of the P4 Header-Female 2.54_1x8.
 - GND is connected to pin 5 of the P4 Header-Female 2.54_1x8.
- Data Connections:**
 - pin8 is connected to pin 1 of the P2 Header-Male 2.54_1x10.
 - pin9 is connected to pin 2 of the P2 Header-Male 2.54_1x10.
 - pin10 is connected to pin 3 of the P2 Header-Male 2.54_1x10.
 - pin11 is connected to pin 4 of the P2 Header-Male 2.54_1x10.
 - pin12 is connected to pin 5 of the P2 Header-Male 2.54_1x10.
 - pin13 is connected to pin 6 of the P2 Header-Male 2.54_1x10.
 - SDA is connected to pin 7 of the P2 Header-Male 2.54_1x10.
 - SCL is connected to pin 8 of the P2 Header-Male 2.54_1x10.
- Terminal Block Connections:**
 - P3 HEADER_6 is connected to pin 1 of the P1 Header-Female 2.54_1x8.
 - P3 HEADER_5 is connected to pin 2 of the P1 Header-Female 2.54_1x8.
 - P3 HEADER_4 is connected to pin 3 of the P1 Header-Female 2.54_1x8.
 - P3 HEADER_3 is connected to pin 4 of the P1 Header-Female 2.54_1x8.
 - P3 HEADER_2 is connected to pin 5 of the P1 Header-Female 2.54_1x8.
 - P3 HEADER_1 is connected to pin 6 of the P1 Header-Female 2.54_1x8.